Backup and Restore

In this section, you will use the **Replicator** utility to test backup and restoration of the **Vault** data. Like all the other components, the **Replicator** utility has already been installed in your environment by the implementation team. It has been installed on the server **comp01**, but to configure and run it, you need to use an administrator account such as *dom*.

Enable the users Backup and Operator

We will be using two **CyberArk** built-in users. The first user is *Backup*, which has permissions to backup all Safes. We will use *Backup* to execute the back up of all Safes. The second user is *Operator*, which has authority to restore all Safes. We will use *Operator* to restore a Safe. These two users are disabled by default, so for this exercise you will need to enable both in your environment in the same way we did for the *Auditor* user in the previous exercise.

Log in to the **PVWA** as *Mike* and enable the users ***Backup*** and ***Operator***. Set both users' passwords to *Cyberark1*.

Create a test Safe and an Account

To test back up and restore, we will create a test Safe and Account, run a replication, delete the Safe, and then restore it from the backup.

Log in as *Mike* and create the following Safe and Account:

Safe name:		*TEST_DELETE*
Account information:	***Platform***	*Lin Ssh 15*
	Name:	*root10b*
	Address	*lin-451.acme.corp*
	Password:	*Cyberark1*

Configure the CyberArk Replicator Utility

Configure the *Vault.ini* file

1. Connect to the server **comp01a** via RDP using the account *dom* / *Cyberark1* (there is a shortcut on the desktop). Once connected, open Windows File Explorer and go to *C:\Program Files (x86)\PrivateArk\Replicate*.

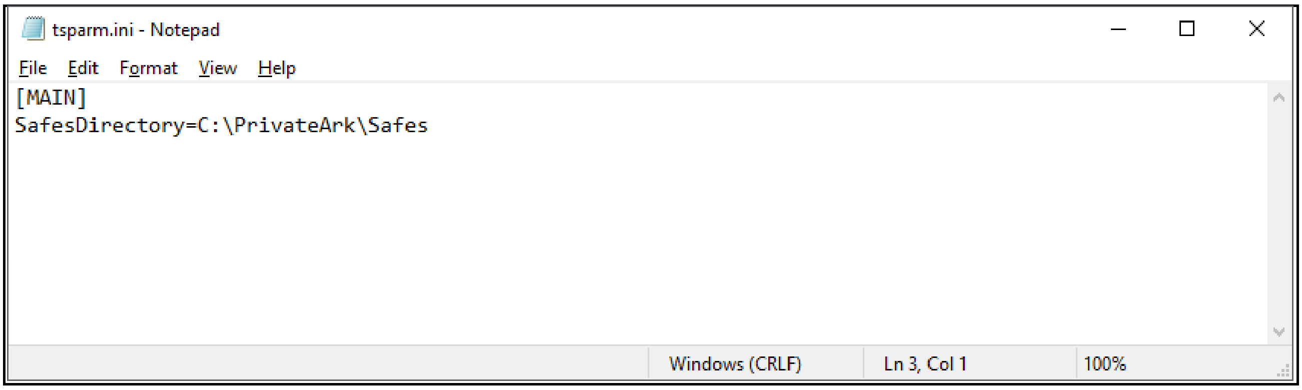
2. Open the *Vault.ini* file, enter "Primary Vault" for the **VAULT** parameter (although this is not mandatory), and enter the IP address of your **Vault** server in the **ADDRESS** parameter: *10.0.10.1*:

```
VAULT = "Primary Vault"
ADDRESS=10.0.10.1
PORT=1858
```

3. Save and close the file.

Locate the output directory – *tsparm.ini*

In the same directory, open the file *tsparm.ini* and note the output location of the backup.



Create the credential file – *backup.cred*

This process is almost identical to the credential file generation that we performed in the preceding exercise. In this case, the *CreateCredFile.exe* is in the same directory as *Vault.ini*.

1. Open a Powershell window (you do not need to do this as an administrator because *Mike* has been given access) and change directories to the *Replicate* folder:

```
cd "c:\Program Files (x86)\PrivateArk\Replicate"
```

2. In this example, we will lock down our credential file by adding additional parameters such as the full path to the executable that is allowed to use the credential file and DPAPI machine protection. Run the following:

```
./CreateCredFile.exe backup.cred Password /username backup /password Cyberark1
/ExePath "C:\Program Files (x86)\PrivateArk\Replicate\PAReplicate.exe"
/IpAddress /Hostname /AppType CABACKUP /EntropyFile /DpapiMachineProtection
```

Note:	You can also find a copy of those commands in the *A1-COMMANDS.txt* file in the *Replicate* home folder.

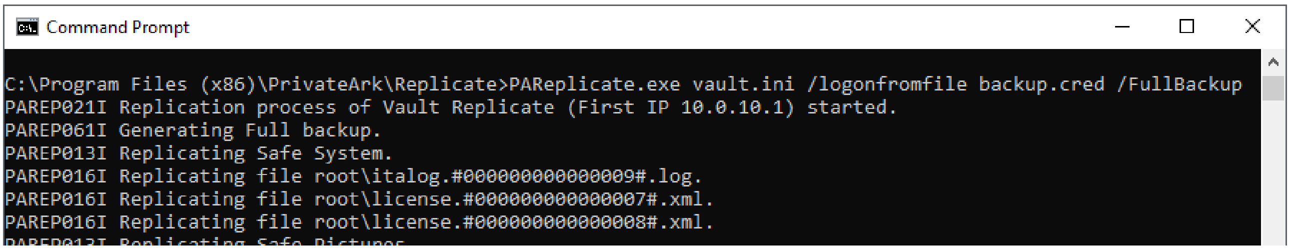
Run a Backup

Note:	**Make sure** that you have enabled the ***Backup*** and ***Operator*** **users** (and set the passwords to ***Cyberark1***) ***prior*** to running the backup command. These users can be found under the System branch. The process is identical to enabling the *Auditor* user performed earlier.

To perform a backup, run the following command from the *Replicate* folder:

```
./PAReplicate.exe vault.ini /logonfromfile backup.cred /FullBackup
```

If the backup is successful, you should see several messages indicating that files are being replicated with a final message stating that the replication process has ended. If everything worked properly, proceed to the next steps. If not, verify the configuration information and try again.



Delete A Safe

In this exercise, we will delete a **Safe** that we will later restore from the backup made in the previous exercise.

1. Login to the **PVWA** as *Mike* and go to **POLICIES | Safes**.

2. Locate the **Safe** TEST_DELETE, click the ellipsis, and then the **Delete** button.

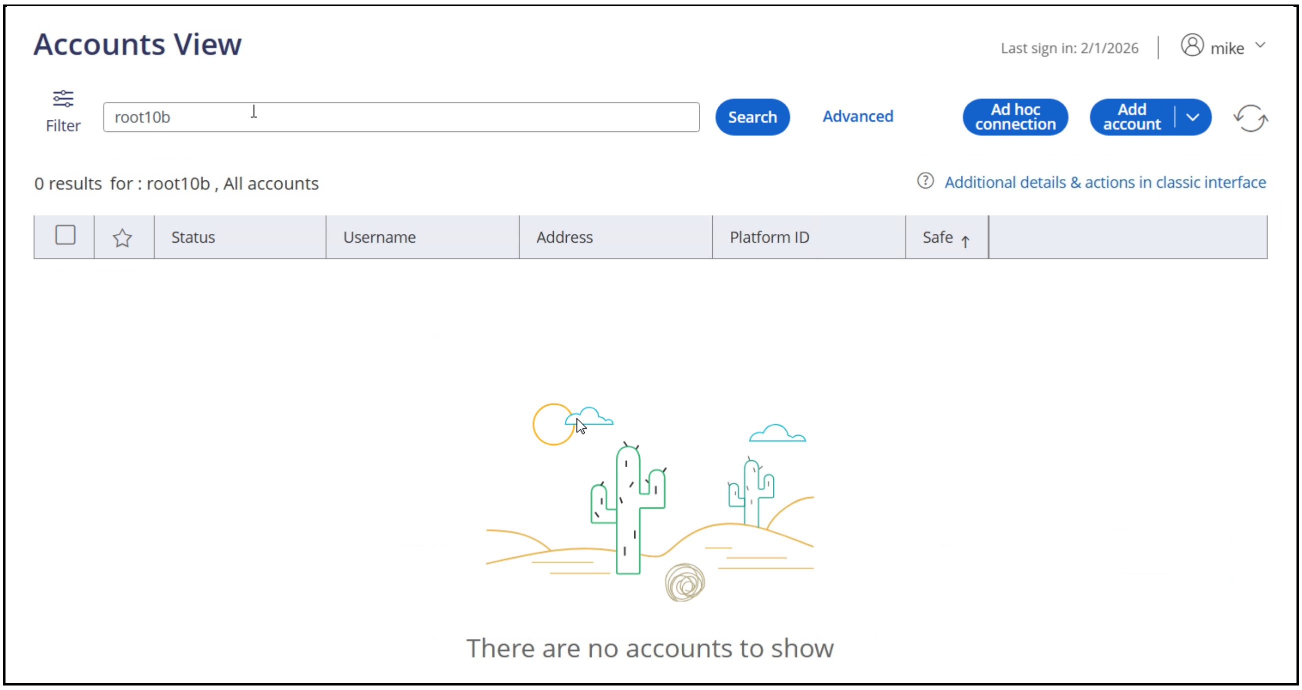
3. You will receive a prompt asking you to confirm deletion. Press **Delete** to confirm that you would like to delete the **Safe** and its contents.

4. Lastly, you will receive a message stating that the **Safe** cannot be deleted due to Safe retention rules. The **Safe** has not been permanently deleted, but it has been removed from usage. Click **Close**.

5. To confirm that the contents of the *TEST_DELETE* **Safe** have been removed, go to the **Accounts** page.

6. Enter *root10b* in the search box and press the **Search** button.

7. The account that you created earlier should not appear.



Run a Restore

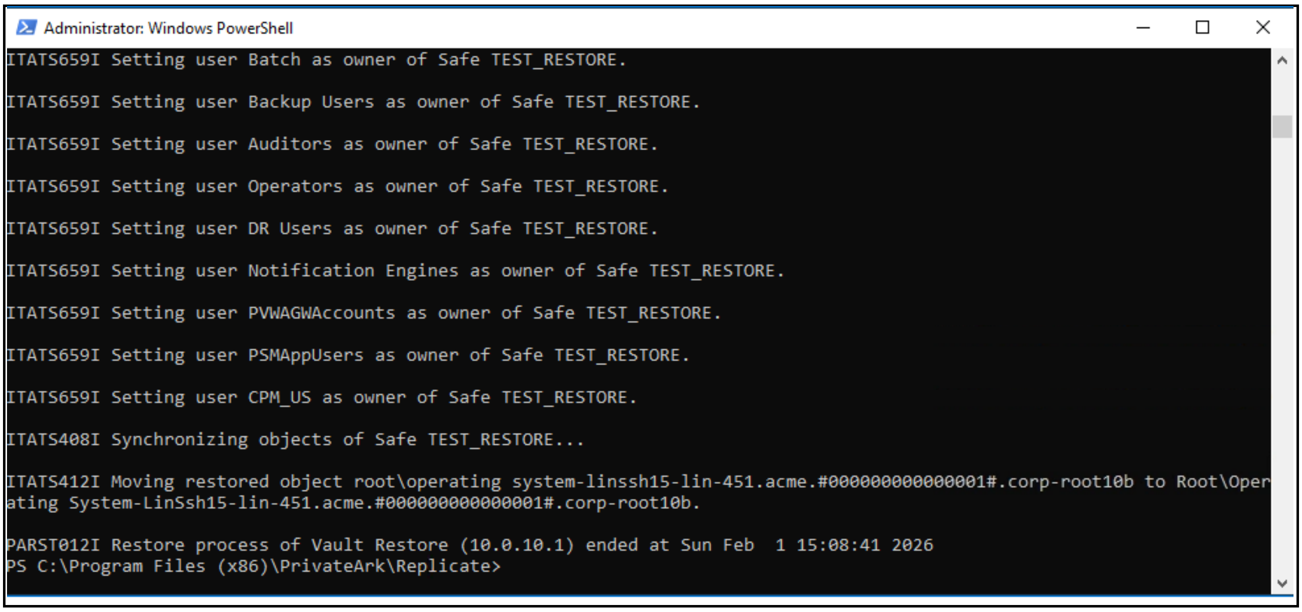
In this section, we will restore the *TEST_DELETE* Safe from the backup we performed earlier. Because the *TEST_DELETE* Safe has not been permanently deleted, we will restore the contents of the Safe to a new Safe named *TEST_RESTORE*.

Note:	Make sure you have enabled the *Operator* user (and set the password to *Cyberark1*) prior to running the restore command. The process is identical to enabling the *Auditor* user performed earlier.
Note:	The *Operator* user is located under the **System** branch.

1. Go back to the command prompt, making sure you are still in the *Replicate* directory, and run the following command:

```
./PARestore.exe vault.ini operator /RestoreSafe TEST_DELETE /TargetSafe
TEST_RESTORE
```

You will be prompted for the password for the *Operator* user, which should be *Cyberark1*.



2. Once you see the message that the restore has been completed, go back to the **PVWA** (as *Mike)* and search for *root10b* again. You should now see the account residing in Safe **TEST_RESTORE**.

Note:	The Target Safe (**TEST_RESTORE**) is the name of the restored **Safe**. The restore process will not overwrite an existing **Safe** (remember, the original has been removed, but not yet deleted) so we must create a new one. Also, it may take a moment for the restored **Safe** to appear, so be patient.